Pressure and Altitude

Student sheet

Name: _____ **Date:** _____

Safety: On the stairs, walk, do not run; hold the box with both hands and read the screen only standing on a landing.

1. What I see on the screen

The screen shows **Pressure** (hPa), **Calc. altitude** (m) and **Temperature**. Move on with the green button at the bottom right; go back with **Back**. **You wait** 1–2 **minutes on every landing**, until the strip turns green (**I moved the box** restarts it), then press **Capture**: the table row on screen fills in. You also write the values here, **with the time**. **The altitude on the card can be hundreds of metres wrong; the difference between two levels is not.**

2. What I do on my own

Building: _____ **Weather today:** _____ **Box:** _____

First I predict. I do not change the prediction after I measure.

1. At the top, the pressure will be ☐ bigger ☐ smaller ☐ the same as at the ground floor, by _____ hPa,
because _____

2. I would know I was wrong if _____

First I measure the stairwell, with the tape measure. I do not guess. One step riser = _____ cm, steps between two floors = _____, one storey = _____ m, storeys climbed = _____, so **the true height** = _____ m

Now I measure. The time is written every time — without it I cannot correct for the weather drift.

Level	**Time**	P **(hPa)**	**App altitude (m)**	T **(°C)**
Ground floor (mean)				
Floor 1				
Floor 2				
Top floor (mean)				
Ground floor (return)				

The check. Two readings at the ground floor and two at the top, 30 s apart; enter the mean in the table.

Ground floor: _____ _____ spread (max − min) = _____ hPa

Top floor: _____ _____ spread = _____ hPa

Difference between levels = _____ hPa. If the spread exceeds it, the difference is **not detectable**, not non-existent: ☐ detectable ☐ not detectable

The return is not optional. Weather drift = (return) − (first reading) = _____ hPa. Do I trust the result? ☐ YES ☐ NO

The rule: for every metre you climb, the pressure falls by 0.12 hPa (that is, 1.2 hPa per 10 m).

Conversion, already worked out: 0.4 hPa → 3 m 0.8 → 7 m 1.2 → 10 m 1.6 → 13 m 2.0 → 17 m

ΔP = ground floor _____ − top _____ = _____ hPa $\Delta h = \Delta P / 0.12$ = _____ m

From the tape measure: _____ m deviation = _____ m as a percentage = deviation / tape measure × 100 = _____ %

The graph. Pressure (hPa) on the vertical axis, height from the tape measure on the horizontal axis. **The pressure axis does not start at zero.**

top: _____

bottom: Height above the ground floor (m), from 0 to _____ — ten divisions

_____

Gradient, from two points *on the line*: _____ hPa/m; theory gives −0.12 hPa/m.

3. The conclusion I write

I predicted that, and the data show that _____

_____

The altitude on screen is wrong, but the difference is good, because _____

_____

What limited me most was _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____